Assignment

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Here is a complete C++ solution tailored to meet all 5 requirements of my assignment, followed by a breakdown to help you with my handwritten explanation.

```
#include <iostream>
#include <string>
#include <vector>

using namespace std;

// Requirement 5: Uses at least one function
// This function determines if a student passes or fails based on
their average.
string checkPassStatus(double average) {
    // Assuming a passing mark average of 50
    if (average >= 50.0) {
        return "Passed";
    } else {
        return "Failed";
    }
}

int main() {
    char choice;

    cout << "=== Student Result Management System ===" << endl;

    // Requirement 4: Uses a loop to allow multiple students to be
entered
    do {
        string name;
        double mark1, mark2, mark3;
        double total;
        double average;

        // Requirement 1: Accepts a student's name and marks for 3
subjects
        cout << "\nEnter Student Name: ";
        cin.ignore(); // Clears the input buffer
        getline(cin, name);

        cout << "Enter marks for Subject 1: ";
        cin >> mark1;
        cout << "Enter marks for Subject 2: ";
        cin >> mark2;
        cout << "Enter marks for Subject 3: ";
        cin >> mark3;
```

```
        // Requirement 2: Calculates the total and average
        total = mark1 + mark2 + mark3;
        average = total / 3.0;

        // Requirement 3 & 5: Displays whether the student
passed/failed using the function
        string status = checkPassStatus(average);

        // Displaying the results
        cout << "\n------ Performance Summary ------" << endl;
        cout << "Student Name: " << name << endl;
        cout << "Total Marks : " << total << endl;
        cout << "Average Mark: " << average << endl;
        cout << "Status      : " << status << endl;
        cout << "-----------------------------------" << endl;

        // Ask the user if they want to continue the loop
        cout << "\nDo you want to enter another student? (y/n): ";
        cin >> choice;

    } while (choice == 'y' || choice == 'Y');

    cout << "\nThank you for using the Student Result Management
System!" << endl;

    return 0;
}
```

📝 Guide for Your Handwritten Explanation

Since your instructor noted they will check your handwritten explanation carefully, here is a structured breakdown of how the code works that you can write down in your own words:

1. Header Files and Setup

- #include <iostream> is used for standard input/output operations (cin and cout).
- #include <string> allows us to use the string data type for the student's name.

2. The Custom Function (checkPassStatus)

- This function takes a double (the average mark) as an input parameter.
- It uses an if-else statement to check if the average is 50 or above. If it is, it returns the text "Passed"; otherwise, it returns "Failed".

3. The do-while Loop (Main Logic)

- A do-while loop is implemented to ensure the program runs at least once, and keeps running as long as the user inputs 'y' or 'Y' at the end.
- Inside the loop, cin.ignore() and getline(cin, name) are used so that names with spaces (like "John Doe") can be read properly without breaking the input stream.
- The total is calculated by adding the three marks (m_1 + m_2 + m_3), and the average is found by dividing the total by 3.
- The program calls the checkPassStatus function, prints the final summary, and asks if you want to repeat the process.

🚀 Next Steps for Your Submission:

1. **Test it on your laptop:** Open your IDE (like VS Code, Code::Blocks, or CLion), paste the code, and run it to ensure it works smoothly before your instructor checks it.
2. **Push to GitHub:** Create a repository, commit this .cpp file, and submit the link before Sunday midnight.
3. **Write it down:** Copy the logic explanation above onto paper clearly for Monday's turn-in.